

Project Report: Molecular Dynamics Simulation of Nanoindentation in Cu (001)

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1. Objective

To computationally simulate the nanoindentation of a pristine Copper (001) single crystal using Molecular Dynamics (MD). The analysis evaluates the elastic-to-plastic transition by identifying the macroscopic yield point (the "pop-in" event) on a force-displacement curve and correlating it directly with the physical nucleation of atomic dislocations.

2. Methodology

The indentation process was modeled in LAMMPS to capture the primary mechanical and atomic responses:

- Interatomic Potential:** The Mishin (2001) Embedded-Atom Method (EAM) alloy potential was selected to accurately reproduce the stacking fault energies and elastic constants of copper.
- Indenter Parameters:** A virtual spherical repulsive force (radius = 30 Å) was applied to the free (001) surface at a constant velocity of 0.1 Å/ps at 300 K.
- Data Analysis & Visualization:** A vectorized MATLAB script was used to plot the raw force-depth outputs. OVITO was used to isolate the resulting defects by applying the Dislocation Extraction Algorithm (DXA) and using Common Neighbor Analysis (CNA) to visually filter out the bulk perfect FCC atoms.

3. Results and Discussion

The computational model successfully generated the force-displacement profile alongside the corresponding atomistic trajectory.

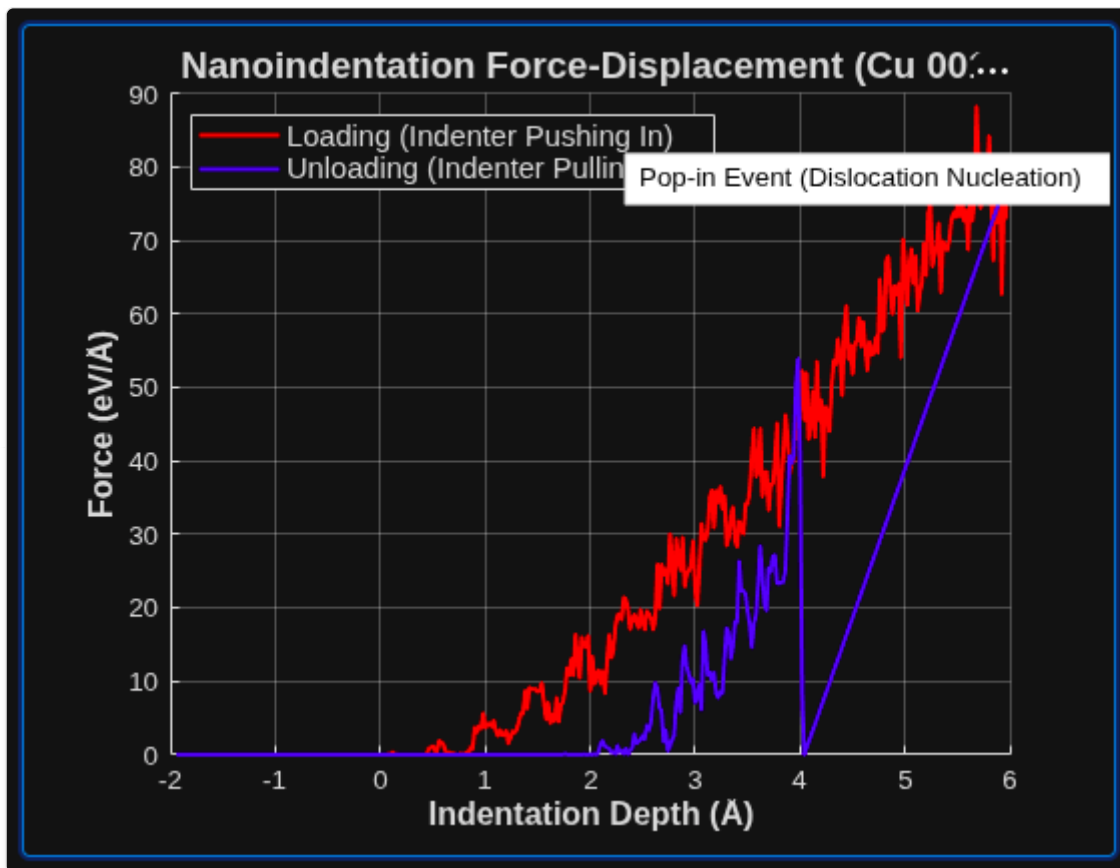


Figure 1: Force-displacement curve for the Cu (001) indentation, showing distinct loading (red) and unloading (blue) phases.

- **Elastic Loading & Pop-In:** During the initial loading phase, the applied force rises smoothly due to Hertzian elastic compression. At an indentation depth of 4.0 Å, the force reaches a critical threshold (approximately 50-55 eV/Å) and drops abruptly to roughly 30 eV/Å. This drop is the "pop-in" event, marking instantaneous stress relief via localized plastic yielding.
- **Unloading (Plastic Deformation):** The unloading curve does not retrace the loading path. Instead, it reaches zero force at an indentation depth of approximately 2.5 Å, confirming that the indenter left a permanent plastic crater in the surface.

By applying a boolean expression (`StructureType == 1`) in OVITO to delete the perfect FCC lattice, the physical mechanisms driving the macroscopic pop-in event become visible:

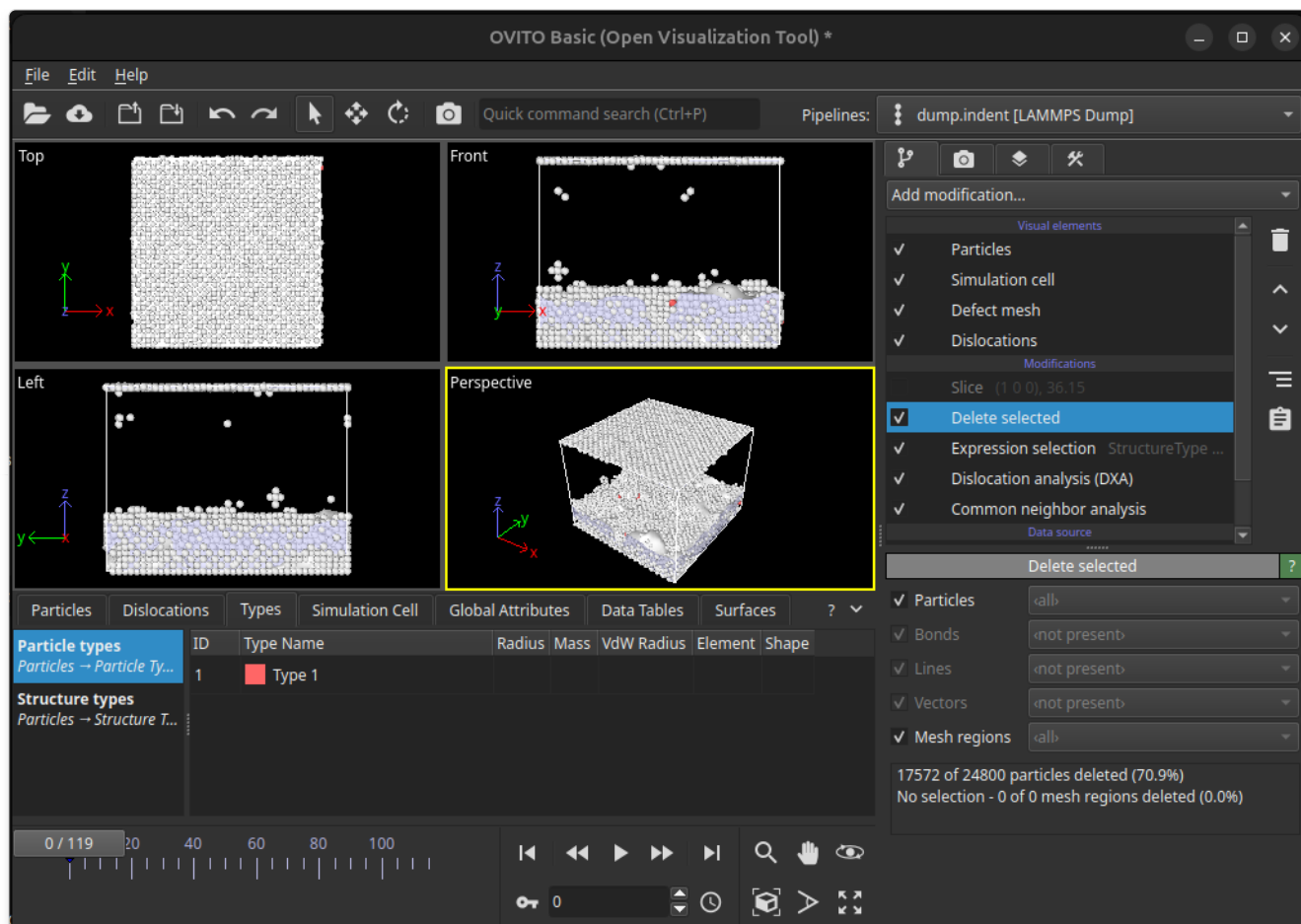


Figure 2: Frame 0 (Pre-Yield). The isolated surface atoms exhibit a perfect, defect-free structure prior to reaching the elastic limit.

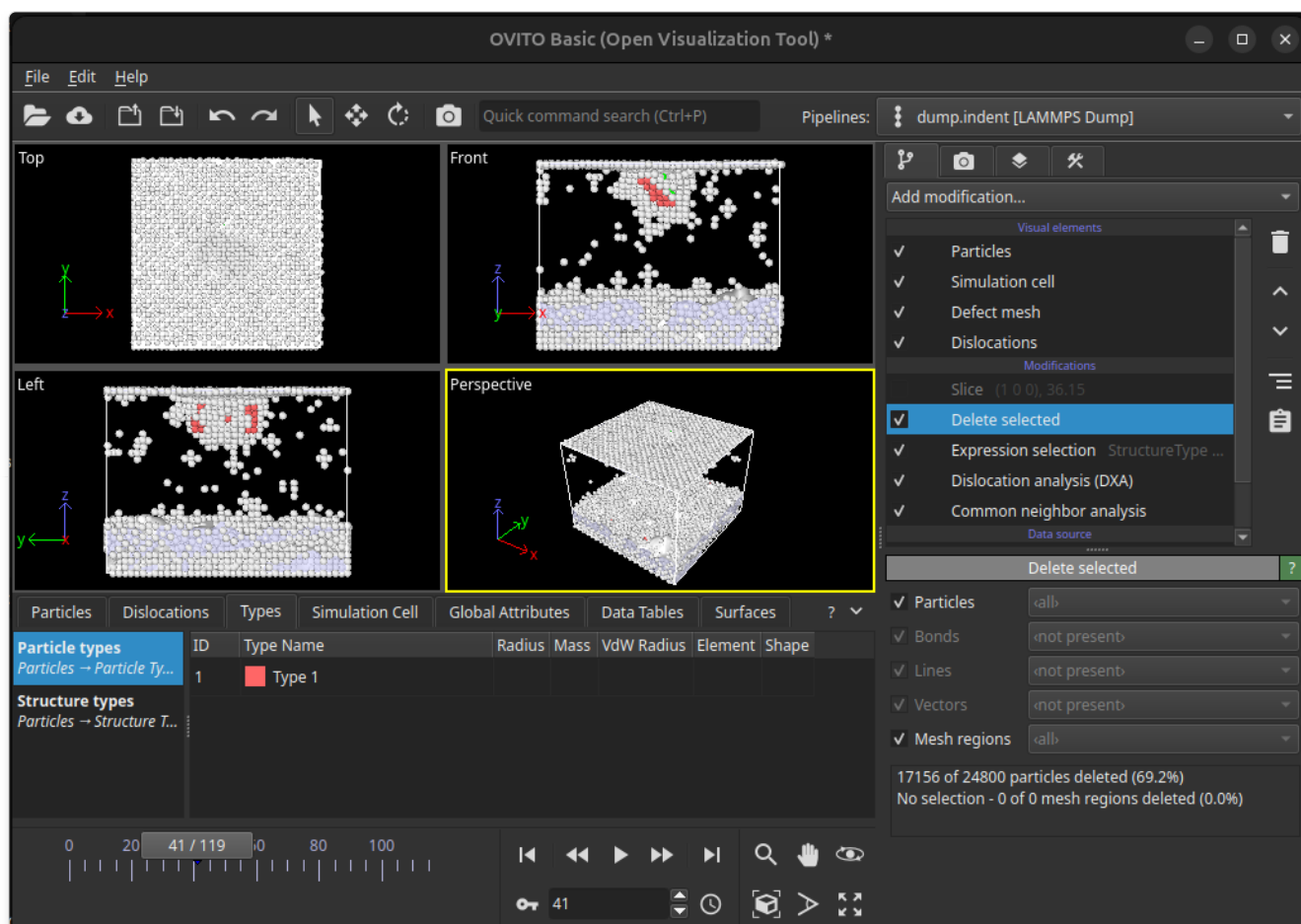


Figure 3: Frame 41 (Onset of Yielding). As the indenter surpasses the critical stress threshold, the initial breakdown of the perfect lattice occurs, with red stacking faults nucleating directly beneath the contact zone.

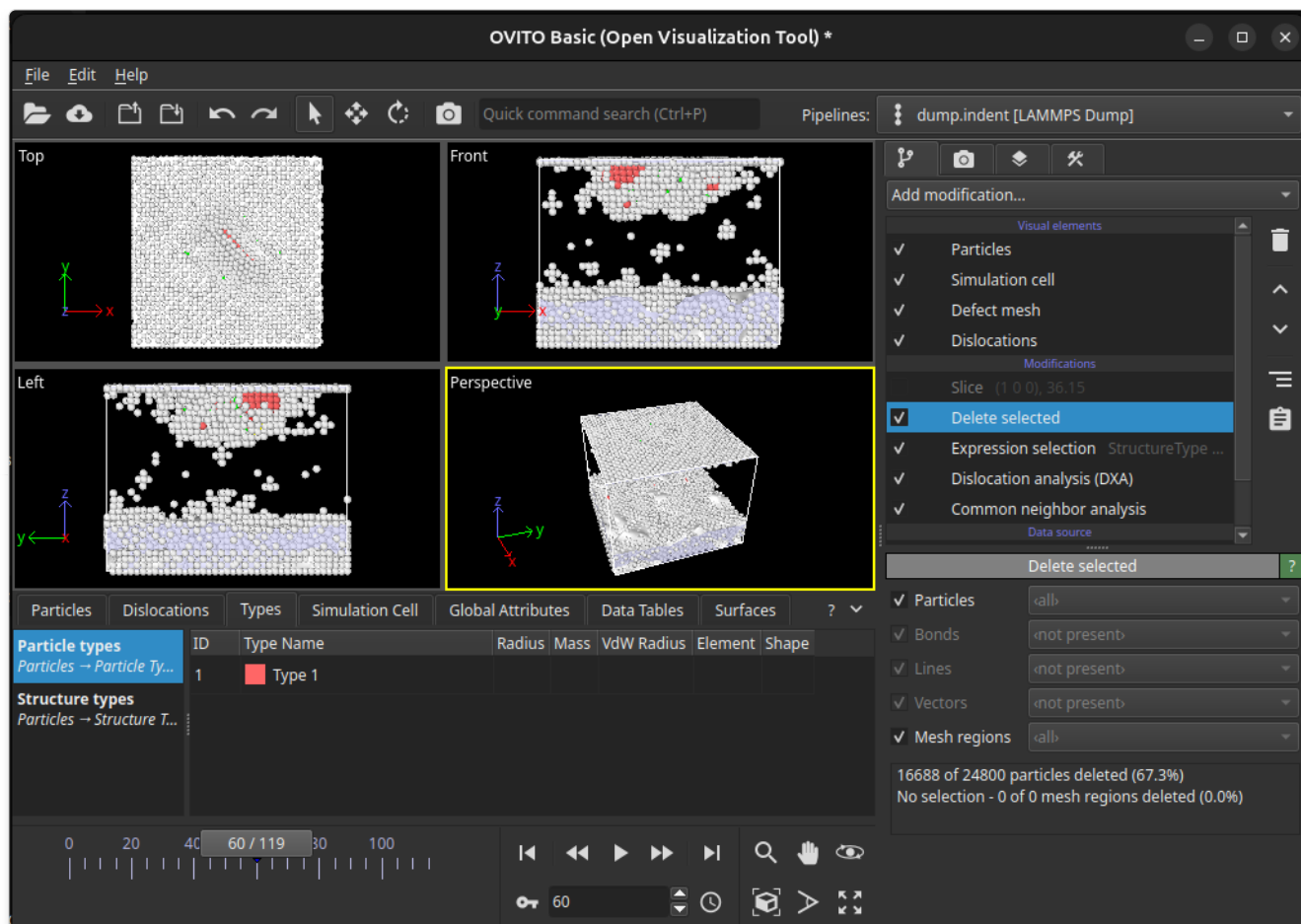


Figure 4: Frame 60 (Dislocation Propagation). Correlating exactly with the force drop at 4.0 Å in Figure 1, a fully formed, V-shaped dislocation loop punches downward into the bulk material.

4. Conclusion

The model demonstrates a direct physical link between continuum contact mechanics and localized atomic defect generation. Because the simulated Cu (001) block contains no pre-existing defects, the discrete pop-in event observed on the force-displacement curve corresponds to true homogeneous dislocation nucleation. The stress required to force these initial defects into the bulk material computationally approaches the theoretical maximum shear strength of a perfect copper crystal.

5. References and Data Sources

1. Mishin, Y., Mehl, M. J., Papaconstantopoulos, D. A., Voter, A. F., & Kress, J. D. (2001). "Structural stability and lattice defects in copper: Ab initio, tight-binding, and embedded-atom calculations." *Physical Review B*, 63(22), 224106.
2. NIST Interatomic Potentials Repository (<https://www.ctcms.nist.gov/potentials/>).
3. Plimpton, S. (1995). "Fast Parallel Algorithms for Short-Range Molecular Dynamics."

Journal of Computational Physics, 117, 1-19. (LAMMPS software framework).

4. Stukowski, A. (2010). "Visualization and analysis of atomistic simulation data with OVITO—the Open Visualization Tool." *Modelling and Simulation in Materials Science and Engineering*, 18(1), 015012.